Against Proto-Algonquian-Blackfoot: How synchronic analysis informs subgrouping

Abstract

This paper focuses on subgrouping within families, which is known to be a difficult problem (Clackson 2022, Harrison 2003, Ringe & Eska 2013a). The empirical question is whether Blackfoot is a sister to Proto-Algonquian, such that the remaining "Core Algonquian" languages form a subgroup. These languages share a phonotactic restriction to the exclusion of Blackfoot: outside of a handful of roots, no words in "Core Algonquian" begin in initial *i. Goddard (2018) argues that this is the result of a shared innovation (the loss of initial *i) in Proto-Algonquian. Based on a synchronic analysis of root alternations, I argue instead that Blackfoot innovated stem-initial *i- in several contexts. This analysis shows that "Core Algonquian" roots, and Blackfoot roots in some contexts, share retentions from a common ancestor, with no evidence for internal subgroups. This work highlights the importance of synchronic analyses and internal reconstructions in subgrouping.

Keywords: Algonquian, Blackfoot, subgrouping, innovations, retentions

Contents

1 Introduction

This paper focuses on the question of subgrouping within historical linguistics, which is known to be difficult (Clackson 2022, Harrison 2003, Ringe & Eska 2013a). A subgroup implies that some subset of related languages has undergone a period of common descent and shared development. The bar for proving a subgroup is necessarily high, as researchers must rule out alternative hypotheses such as parallel development or retention from a higher-level proto-language. This paper examines one proposal for subgrouping in detail, ultimately arguing that an internal reconstruction supports a retention analysis rather than an innovation.

The empirical context is the position of Blackfoot (ISO-639-3: bla) within the Algonquian family. Blackfoot has long been labelled "divergent" with respect to the rest of the family (e.g. Bloomfield 1946: 85). Not only does it exhibit a number of innovations in the lexicon (Voegelin 1941: 506), phonology (Goddard 1994: 187), and inflectional system (Goddard 2018) which obscure cognates, but it also contains forms with archaic meanings which have since been lost (e.g. Goddard 1994). Because of these facts, some researchers have argued that Blackfoot is a sister to the other Algonquian languages. This was first suggested by Proulx (1980: 14) and later argued more strongly by Goddard (2018). This analysis is illustrated in (1), where I follow Goddard (2018) in labelling the non-Blackfoot languages as "Core Algonquian". The traditional subgrouping, (2), leaves Blackfoot as a sister to the other languages in a flat structure (Goddard 1994, Michelson 1935, Valentine 2001).

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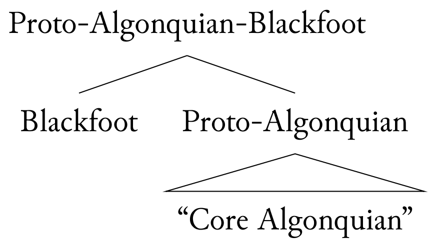

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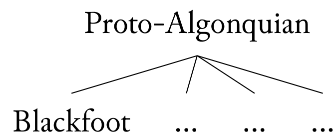


The tree in (1) predicts that Core Algonquian should form a subgroup; that is, that they should have shared innovations which exclude Blackfoot. Goddard (2018) presents one phonological innovation and one morphological innovation supporting his argument that "Core Algonquian" forms a subgroup. Since phonological sound changes are still the "gold standard" for subgrouping, I focus on the phonological claim, which is that stem-initial *i- was lost in Proto-Algonquian (the ancestor to "Core Algonquian") but retained in Blackfoot. I argue instead that that a synchronic morphophonological analysis of Blackfoot and subsequent internal reconstruction leads to a simpler analysis, where "Core Algonquian" shares retentions from a shared ancestor, and Blackfoot innovated stem-initial *i- is several contexts. This analysis points only to a shared ancestor, in favor of (2), and highlights the importance of synchronic morphophonological analyses and internal reconstructions in subgrouping.

The paper is structured as follows. In Section 2 I review the difficulties of determining subgroups within language families. In Section 3 I discuss Blackfoot's contested position in Algonquian and explain why the question of how Blackfoot relates to the other languages is interesting. Section 4 presents a basic phonological background of Proto-Algonquian and Blackfoot. Section 5 presents an overview of Goddard's (2018) phonological argument for subgrouping. In Section 6 I argue that there is no strong evidence that there was a single phonological innovation in Proto-Algonquian rather than multiple parallel developments in each "Core Algonquian" language. Then in Section 7 I present an alternative analysis where the "Core Algonquian" languages retain an earlier pattern and instead Blackfoot contains several phonological innovations. Section 8 contains an discussion, and Section 9 concludes.

2 The difficulty of subgrouping

Subgrouping implies a period of shared development or common descent within a language family; that is, that some set of languages only diverged at a later stage in their development. Harrison (2003) discusses subgrouping as one of the limits of the comparative method. Whereas the comparative method establishes genetic relatedness among a set of languages, subgrouping is a "tree selection problem" that seeks the best representation of speciation within that set of languages (Harrison 2003: 232–233). In other words, subgrouping finds the most economical relative chronology of shared sound changes.

It has been recognized since Brugmann (1884) that periods of shared development can only be established on the basis of shared innovations rather than shared retentions (e.g., Crowley 1997: 167, Hoenigswald 1960: 151, Harrison 2003, Ringe & Eska 2013a).[1] Crucially, in order for a shared feature to be an innovation, it must *not* be due to (i) parallel development (or drift), (ii) diffusion across language boundaries, or (iii) shared retentions. This means that the work of the linguist is to avoid "false positives" for innovations, which in practice can be difficult.

In parallel development, a sound change occurs in each language individually at some point later than the period of shared development (e.g., after the languages had already diverged). It is also possible for a sound change that begins during a period of shared development to continue to completion in some or all of the various daughter languages after the languages have diverged. This is far more likely to occur with typologically common or phonetically natural sound changes. Shared sound changes in Algonquian are often attributed to diffusion across language boundaries, though often without strong evidence (see Salmons 2025 for discussion). If the shared sound changes are common or natural, then the patterns from diffusion may look identical to parallel development. The exception is when the languages that share a linguistic feature are not (and were not) geographically contiguous, in which case diffusion due to language contact is not possible (Harrison 2003: 236). Finally, the null hypothesis is that a shared feature among a set of languages is a retention from an earlier state. In this case, the other languages in the family do not share the feature either because they underwent separate innovations, lost the retention, or both.

What are the best types of evidence for a shared innovation rather than one of these false positives? It can be difficult to tell whether vocabulary replacement and syntactic change constitutes an innovation or not (see examples in Clackson 2022, Harrison 2003). In contrast, a partial or full phonological merger cannot then be later undone, so phonological sound changes are the clearest indicators of innovation (Hoenigswald 1960: 151). Typically, researchers look for uncommon or unusual sound changes, or sets of several phonological changes which would not ordinarily be expected to have taken place together (Crowley 1997: 169, Harrison 2003: 234). Uncommon or usual sound changes these are less likely to be explained by parallel development, which may occur due to extrinsic phonetic pressures or common processes of semantic

[1]Although this point apparently often needs to be reiterated (e.g., Koch & Bowern 2004, Ringe 2017: 63).

development or grammaticalization. Even common or natural sound changes may be taken as evidence of an innovation if they are sporadic or irregular in some way, such as occurring across a random set of lexical items. A randomly distributed sound change is unlikely to be repeated in multiple languages by change. Barring that, even a common or natural sound change could support a subgroup argument if it is one of many shared innovations among that set of languages. In some cases, the relative chronology of sound changes in different languages can be used to determine which shared developments are innovations and which are not (Meillet 1908: 10). However, the relative chronology of sound changes is often underdetermined.

It practice then, it can be difficult to argue for subgrouping because it is difficult to argue that any particular change is an innovation. In the next section I describe the problem of subgrouping as it relates to Blackfoot and the Algonquian family.

3 Blackfoot's relation to Algonquian

Blackfoot is geographically and phonologically distant to the other Algonquian languages (Section 3.1). Many cognates have now been identified or claimed with the other languages, and it is clear that Blackfoot has undergone significant phonological innovations (Section 3.2). This has led to multiple claims about Blackfoot's position within the Algonquian family, culminating most recently in an argument that Blackfoot is a sister to the rest of the family (Section 3.3).

3.1 Geographic position

The Algonquian language family stretches across North America from the Atlantic Ocean to the foothills of the Rocky Mountains (Frantz 2017, Goddard 1975, Mithun 1999: 336–337). Blackfoot (outlined in black in Figure 1) is the westernmost language within this family, and is spoken in southern Alberta, Canada and northern Montana, USA (Frantz 2017).

Known as Niitsí'powahsin ('real/true language') in Blackfoot itself, it is the language of the Blackfoot Confederacy (Siksikaiitapi), which is an alliance of solidarity among four distinct sovereign Nations that share a common language, culture, and heritage (Dempsey 2019-07-18; Grinnell 1892: 153; Hungry Wolf & Hungry Wolf 1989: 3; Juneau 2007: 13*ff*). The location of the reserves and reservations associated with these Nations are shown in gray in Figure 2. The Blackfoot and English names of the three Nations in Canada are Siksiká (Blackfoot), Káínai (Blood), Aapátohsipikani (Peigan, or Northern Peigan). The fourth Nation, Aamsskáápipikani or Piikúnni (Blackfeet, or Southern Piegan),[2] is located in the USA. For recent descriptions of the numbers of speakers and dialects, see Genee & Junker (2018) and Weber et al. (2023).

3.2 Innovations and divergence

Perhaps because Blackfoot is geographically peripheral, it is linguistically quite different from the rest of the family. Researchers have long noted many lexical differences (Voegelin 1941: 506), phonological innovations (Goddard 1994: 187), and inflectional innovations (Goddard 2018). The many innovative sound changes "all contribute towards making Blackfoot vocabulary as a whole appear as un-Algonquian" (Michelson 1935: 142–143), and contribute to the difficulty of

[2]The spelling is usually 'Peigan' in Canada, but 'Piegan' in the United States.

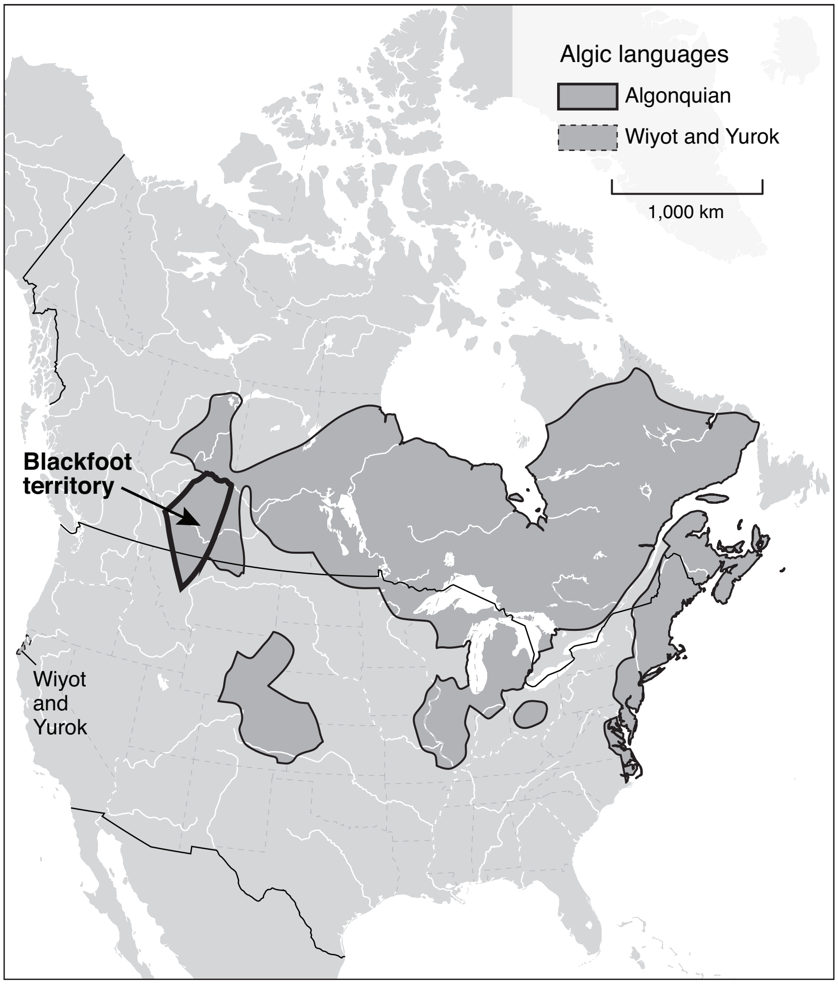


Figure 1: Algonquian language family. Map by Eric Leinberger, based on Goddard (1999).

finding likely cognates. Because of this, Blackfoot is frequently labelled as "the most divergent language in the Algonquian family" (e.g. Goddard 2018: 83).

Early explorers and scholars did not believe that Blackfoot was even Algonquian. Hayden (1863: 253) quotes Mackenzie's "General History of the Fur Trade" [likely Mackenzie 1801] as claiming that the Blackfoot "speak a language of their own... nor have I heard of any Indians with whose language that which they speak has any affinity". Similar views are espoused in Franklin (1823: 109), Gallatin (1836: 132), and Howse (1849: 113). Gallatin (1848) was the first to lay out a significant number of cognates to other Algonquian languages, while Michelson (1912), Voegelin (1941) showed that the inflectional system was very similar to other Algonquian languages, albeit with some Blackfoot-specific innovations. Although this early work was instrumental in proving that Blackfoot was Algonquian, it is notable that none of these papers were based on systematic sound correspondences. In fact, Goddard points out that Michelson (1912) "had no way of distinguishing between accidental likenesses, shared innovations, and shared retentions" (Goddard 1967b: 7).

More recent papers on Blackfoot historical phonology include Berman (2006, 2007), Proulx (1989, 2005), Taylor (1960, 1967), Thomson (1978), with Berman (2006) and Proulx (1989)

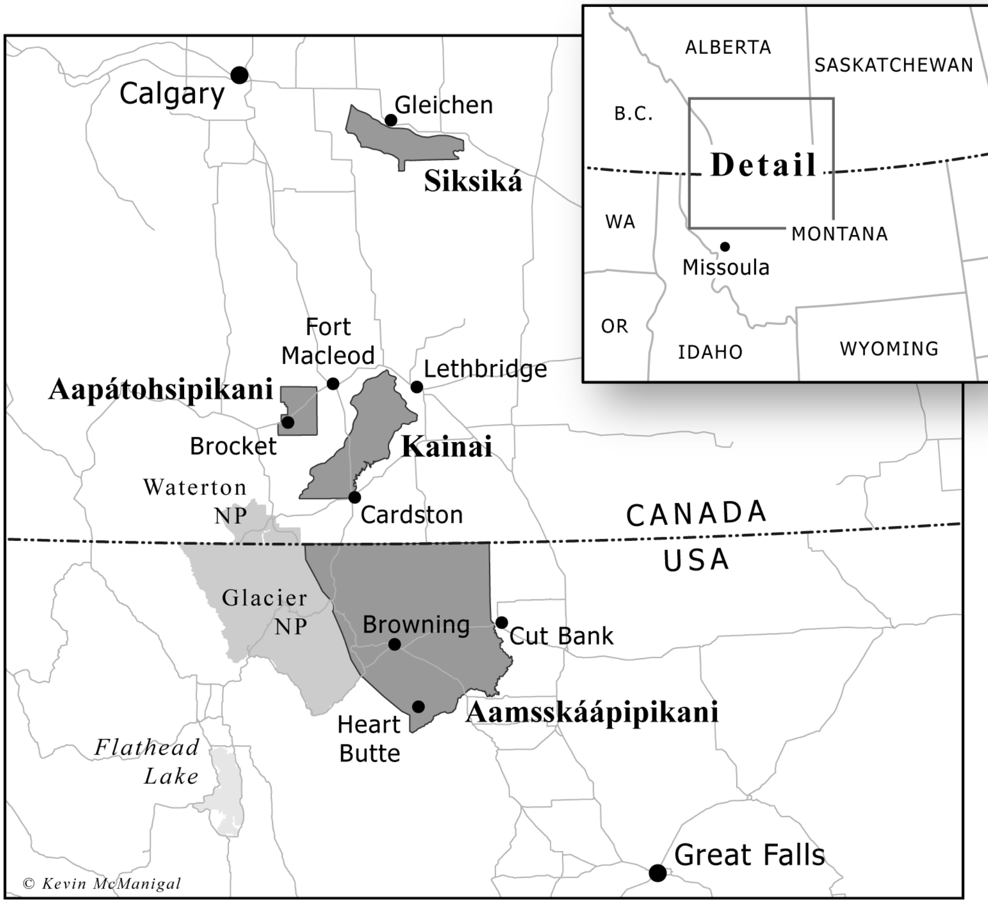


Figure 2: Blackfoot reserves and dialects. Map by Kevin McManigal.

being the most comprehensive of this set. Even so, the historical phonology of Blackfoot remains not well understood. Many of the cognates and reconstructions rely on irregular and unusual sound changes, ranging from the mundane to the highly unusual, such as the rampant metathesis that was once proposed in earlier studies (Michelson 1935: 142, Sapir 1913: 624, 630, Uhlenbeck 1925). Even fairly recent studies propose a number of unusual sound changes, often based only on one or two forms (Berman 2006, Proulx 1989).

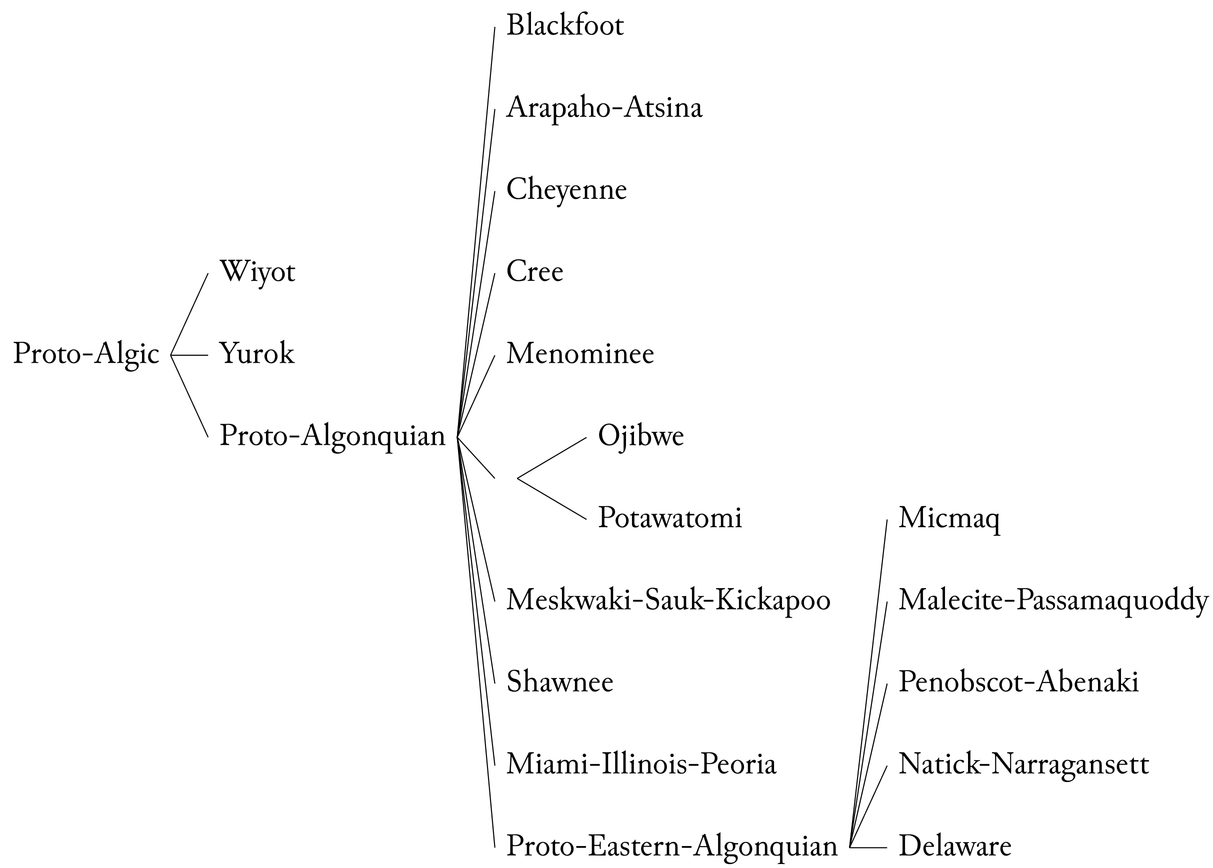
The recent claims that Blackfoot branched first from the family tree (Goddard 1994, 2018, Proulx 1980) must be understood in the context of these earlier claims and of the unusual properties of Blackfoot, relevant to the remainder of the family. I turn to these claims next.

3.3 Cladistic position

The Proto-Algonquian family is well-established via reconstructions of the phonological system (Bloomfield 1925, 1946, Goddard 1979, 1994, Michelson 1920, 1935, Pentland 1979, 2023, Siebert Jr 1941, 1975) and morphological system (Goddard 1967a, 2007, Pentland 1999). The languages are traditionally divided into three groupings (cf. Mithun 1999): Plains Algonquian, Central Algonquian, and Eastern Algonquian. Plains Algonquian and Central Algonquian are areal groups, and not genetic subgroups. Blackfoot is part of the Plains Algonquian group, along with Arapaho and Cheyenne. The Central Algonquian group includes Cree-Menominee, Ojibwe-Potawatomi, Meskwaki, Shawnee, and Miami-Illinois. The only established subgroup in the family is Eastern Algonquian (Goddard 1974, 1980), though even this has been contested (Pentland 1992, 1979, Proulx 1984a). Since Goddard (1994), the consensus has been that Algonquian is structured as a dialect continuum, with the oldest layers in the west and the more recent in the east.

Because of this, the traditional analysis of the Proto-Algonquian family, is a rather flat structure outside of the Proto-Eastern Algonquian subgroup. The tree in (3) in based on Valentine (2001: 14). More recently, Rhodes (2021) and Biedny et al. (2022) converge on a proposal for a Core Central Algonquian subgroup that contains Ojibwe-Potawatomi, Meskwaki-Sauk-Kickapoo, Shawnee, and Miami-Illinois-Peoria. See Salmons (2025) for an overview of other subgroup proposals in Algonquian.

(3)



The "divergent" nature of Blackfoot has led to various claims about how the language is positioned within the Algonquian family. Michelson (1912: 229) put Blackfoot on a branch of its own, and also noted that "though Blackfoot must be classed apart from Eastern-Central Algonquian, it has the closest affinities to Fox, Eastern Algonquian, and Cree" (Michelson 1912: 232). Other scholars have grouped Blackfoot with Cree and Cheyenne (Hayden 1863) or with Arapaho-Atsina and Cheyenne as part of a classification which differs from that in (3) (Pentland 1979). Rather than attempting to group Blackfoot with other languages, Proulx (1980: 14) focused on how divergent Blackfoot is from the other languages and noted that this could either be due to a long history of relative isolation or to a cladistic arrangement where Blackfoot is a sister to Proto-Algonquian. More recently, Goddard (1994, 2018) brings forward more data to essentially argue for each of these possibilities in turn. (Goddard 1994) argues that Blackfoot is the oldest dialectal layer in Algonquian, based on the number of striking phonological innovations as well as "some apparent lexical archaisms in Blackfoot" (Goddard 1994: 188). Goddard (2018) then specifically argues that Blackfoot branched from Porto-Algonquian-Blackfoot, the immediate ancestor to Blackfoot and a new Proto-Algonquian subgroup.

These claims have potential impacts on the reconstruction and subgrouping of Proto-Algonquian. First, Blackfoot potentially maintains contrasts that were lost in the "Core Algonquian" languages and which could alter the Proto-Algonquian-Blackfoot reconstructions. This is similar to the impact that the Eastern Algonquian languages had on Proto-Algonquian reconstructions (see for example, Siebert Jr 1941, who argued that correspondences with the Eastern Algonquian languages revealed an additional cluster in Proto-Algonquian). Second, even if Blackfoot should be positioned elsewhere in the family tree, evidence from Blackfoot could still reveal something new about the internal subgrouping of Algonquian, perhaps adjudicating between the multiple hypotheses mentioned above. However, both of these options require more knowledge of Blackfoot synchronic and historical phonology.

In the next section I lay out the phonological inventories and relevant phonotactics in Proto-Algonquian and Blackfoot. After this, I present the phonological claim in Goddard (2018) in more detail.

4 Phonological background

The presence or absence of word-initial *i in both Proto-Algonquian and Blackfoot is partially conditioned by the following consonant(s). This section therefore presents the consonant and vowel inventories for Proto-Algonquian and Blackfoot, as well as allowable clusters and other phonotactic details.

4.1 Proto-Algonquian

The consonant inventory is given in Table 1, using the same symbols as Goddard (1994, 2018).[3] The glottal stop *ʔ occurred only as the first consonant in a true consonant cluster (i.e., one where the second consonant is not a semivowel). The phonetic value of *θ is uncertain, and many authors argue that this sound was phonetically a lateral fricative (Blevins 2004: 296–297, Picard 1984, 1994, Proulx 1984b: 423, Siebert Jr 1975: 300). I follow Goddard (1994: 204), who concludes that this sound probably was an interdental fricative. The post-anterior affricate *č is in parentheses because its phonemic status is questionable. It is nearly always a predictable allophone of *t before *i, *i·, or *y (Bloomfield 1925: 144, 1946: 92, Goddard 1979: 72), with only a few exceptions in onomatopoeic words (see Goddard 1979: 74).[4]

[3]There are several different transcription systems in use for Proto-Algonquian. Goddard (1994) argues that the transcription system from Bloomfield (1946) should be modified in several ways: he uses *ʔ for *q, *r for *l, *sC for *xC, and *rC for *çC. Other researchers use *ɬ or *ł in place of *θ, or use *sC in place of Bloomfield's *çC.

[4]Many researchers therefore take *č to be non-phonemic in Proto-Algonquian (Hockett 1956, Kaye 1978: 144, Pentland 1979: 340–341, 1983, 1999: 226, 2023: *xiii*). Goddard continues to transcribe *č in reconstructions

Table 1: Proto-Algonquian consonants

	Labial	Coronal		Dorsal	Glottal
		Anterior	Post-anterior		
Stops	*p	*t	(*č)	*k	*ʔ
Fricatives		*θ, *s	*š		*h
Liquids		*r			
Nasals	*m	*n			
Semivowels	*w		*y	(*w)	

Clusters of true consonants (a class which excludes the semivowels) in Proto-Algonquian are given in Table 2. The first member of the cluster is listed down the first column; the second member of the cluster is listed across the top row. Allowable clusters are marked by ✓ or "1"; the latter is used for clusters that are reconstructed on the basis of a single reconstruction. There is one additional cluster *hm which mainly occurs in certain inflectional suffixes (see Goddard 2007: 209 [note 3], 224*ff*, 249–250 for discussion). Bloomfield (1946) additionally reconstructs the clusters *čp and *čk to account for a few reflexes in Menominee, but Pentland (1999: 382–384) argues that these are innovations in Menominee alone and they are not included in Table 2. All consonants (and consonant clusters) could be followed by a semivowel *w or *y, except that *(C)ty, *(C)θy, and *(C)čw did not exist. This is due to a regular process of alternation in Proto-Algonquian, where the sounds *t and *θ are replaced by *č and *š, respectively, before *i, *i·, or *y (Bloomfield 1925: 144, 1946: 92, Goddard 1979: 72).

Table 2: Proto-Algonquian clusters

C1\C2	p	k	t	(č)	š	θ	r	s
m/n	✓	✓	✓	✓	✓	✓	✓	✓
h	✓	✓	✓	✓	✓	✓	1	✓
ʔ	–	–	✓	✓	✓	✓	✓	✓
s	✓	✓	–	–	–	–	–	–
θ	✓	✓	–	–	–	–	–	–
š	✓	✓	1	–	–	–	–	–
r	–	✓	–	–	–	–	–	–

The Proto-Algonquian vowel system, shown in Table 3, contains four vowel qualities and a length contrast (Bloomfield 1946, Goddard 1979, Pentland 1979). Vowel length is indicated throughout this paper by a middle dot (·). Short *o was somewhat marginal and a recent development from pre-Proto-Algonquian *we (Berman 1982: 414, Goddard 1979: 75). Because of this, it is often unclear whether this vowel should be reconstructed as *o or *we in

because many daughter languages develop a phonemic contrast between [t] and [č] (Goddard 1979: 74, Hockett 1956), and *t may be optionally replaced by *č (and *s by *š) in Proto-Algonquian to form diminutives as form of sound symbolism (Goddard 1977: 241, 244, Pentland 1975: 241–248, Teeter 1959: 42–43).

Proto-Algonquian itself, but most researchers follow Bloomfield's (1946) practice of using *we. Goddard has more recently reverted to reconstructing *o and treating this vowel as phonemic in Proto-Algonquian (Goddard 2002: 45, note 2; 2015: 399, note 2; 2018: 102, note 2).

Table 3: Proto-Algonquian vowels

	Front	Back
High	i· i	o· o
Low	e· e	a· a

The contrast between short *i and *e is neutralized in word-initial syllables (e.g., Pentland 1979: 403, 2023: 1418). In many languages this neutralized vowel is phonetically high, [i], in absolute word-initial position, but phonetically mid or lower, [e], after a word-initial consonant. This distribution occurs in Meskwaki, pre-Cheyenne, and Proto-Arapaho-Atsina (Oxford 2015: 320–321) as well as pre-Blackfoot (Berman 2006: 266, Goddard 1994: 192, Proulx 1989: 55, 68–69). Other languages neutralize to [i] in both positions, or [e] in both positions (see discussion in Oxford 2015: 320–321). Because there is no contrast, many researchers reconstruct this vowel as *e regardless of whether the first syllable begins with an onset or not (Bloomfield 1946: 93; Goddard 1979: 72; Pentland 2023: *xiii*). However, others have taken to using *i in absolute word-initial position and *e after a word-initial consonant (see discussion in Oxford 2015: 320–321, and notes in Goddard 2002: 45, note 2; 2015: 399, note 2; 2018: 102, note 2). This is the convention adopted in Goddard (2018) and in this paper.

4.2 Blackfoot

The inventories in Table 4 and Table 5 follow Weber & Derrick (2025) and use the orthography developed in Frantz (1978, 2017). Phonemic values are given in slashes where these differ from the orthography. Blackfoot neutralizes many of the consonants from Proto-Algonquian but has developed a contrast in length for many consonants, indicated by doubled letters. The only consonants without a length contrast are the glides, *h* /x/, and *'* /ʔ/ (Table 4). A long dorsal affricate *kks* occurs only as an allophone of geminate *kk*. Some segments have an unusually restricted or expansive distribution. The dorsal fricative *h* only occurs before obstruents (Reis Silva 2008). The glottal stop *'* typically occurs before a consonant and very rarely between vowels (Peterson 2004). It also occurs predictably in word-initial position, where it is not written (Weber & Derrick 2025). Short and long /s/ occur before, between, and following consonants, where they pattern with vowels (Denzer-King 2009, 2012, Derrick 2007, Goad & Shimada 2014a,b, Weber & Derrick 2025). Note that most of the Proto-Algonquian coronal consonants merged by position, so that *θ and *r become *s* in word-initial position and *t* elsewhere (Berman 2006: 265, AUTHOR 2026b).

There are very few true consonant clusters in Blackfoot: *ssC*, *hC*, and *'C*. The consonant following *ss* or *h* in these clusters is always an obstruent. The consonant after a glottal stop may be an obstruent or a nasal. The second consonant in true consonant clusters is always short. The synchronic motivation for treating these consonant sequences as true consonant clusters is that vowels are shortened before these clusters as well as before geminates (Elfner 2006, Frantz

Table 4: Blackfoot consonant inventory

	Labial	Coronal	Dorsal	Glottal
Stops	p pp	t tt	k kk	' /ʔ/
Pre-Assibilants		st stt		
Affricates		ts tts	ks	
Fricatives		s ss	h /x/	
Nasals	m mm	n nn		
Glides	w	y /j/	(w)	

2017, Weber 2020, Weber & Derrick 2025). The pre-assibilants and affricates in Blackfoot are not clusters. The main reasons are that they are significantly shorter than clusters and that they do not cause preceding long vowels to shorten like true clusters (see also Elfner 2006, Weber 2020, Weber & Derrick 2025). There are also alternations between plain and pre-assibilated or affricate consonants (Chávez Peón 2015, Weber 2020: 36, Weber & Derrick 2025). Clusters and single consonants (except glides and *h*, and for some speakers also the assibilants) can be followed by *y* (but never *w*).

There are three short vowels and five long vowels in Blackfoot (Table 5). Following arguments in AUTHOR (2026a), Weber & Derrick (2025), Weber & Miyashita (2024), I treat some instances of tautomorphemic /ɛː/ and /ɔː/ as phonemic, although the majority of <ai> and <ao> sequences arise across morpheme boundaries in Blackfoot. Doubled vowels usually indicate vowel length except for long mid vowels, which are written with a digraph even when phonologically shortened, and any vowel before *hC clusters, where doubled letters indicate voicing. Note that Proto-Algonquian short *e and *i merged fully in Blackfoot to short *i* (Berman 2006: 266, Oxford 2015: 341).

Table 5: Blackfoot vowel inventory

	front		back
high	i ii		o oo
mid	ai /ɛː/		ao /ɔː/
low		a aa	

The following section lays out the proposal for an innovation in Proto-Algonquian from Goddard (2018). After that, in Section 6 I argue that there is no strong evidence against an alternative hypothesis where the loss of initial short *i occurred independently in each Algonquian language. Then in Section 7 I argue for an alternative hypothesis where the "Core Algonquian" languages do not form a subgroup, and instead only Blackfoot contains innovations.

5 Proposal: "Core Algonquian" languages form a subgroup

Initial short *i in the "Core Algonquian" languages is highly restricted and only occurs in a small number of roots before a limited set of consonants and clusters. In contrast, Blackfoot apparently has many roots which begin in a short *i* vowel. To explain this difference, Goddard (2018: 100) argues that Blackfoot retains an earlier pattern, and that the innovation was in Proto-Algonquian (the ancestor to the "Core Algonquian" languages), which lost most initial *i by regular sound change besides a handful of exceptions. Goddard takes this innovation as partial evidence for a subgroup which excludes Blackfoot. The remainder of this section lays out his proposal, and in some cases expands upon it to add more detail.

5.1 "Core Algonquian" data

As discussed in Section 4.1, the contrast between short *i and *e is neutralized in initial syllables in the "Core Algonquian" languages. Relevant to the subgrouping argument in Goddard (2018), word-initial short *i is even further restricted in Proto-Algonquian (Goddard 2002: 58, 2018: 99). It occurs only in the following three distinct contexts. I discuss the demonstrative and relative roots in more detail below.

1. before demonstrative roots
2. before three relative roots or stems
3. in other roots before
   (a) a single consonant *θ or *r
   (b) before a restricted set of consonant clusters (*hk, *šk, *šp, *rk, *θk)

The demonstratives have developed in different ways in each of the daughter languages, but a core set of roots can be reconstructed back to Proto-Algonquian. There are two proposals for the the historical development of Proto-Algonquian demonstratives (Goddard 2003, Proulx 1988), which differ in whether they reconstruct an initial *i back to Proto-Algonquian. Goddard (2003) reconstructs multiple series of demonstratives, named simply "Set A" through "Set F", on the basis of Meskwaki-Kickapoo, Shawnee, and Eastern Algonquian. There is a basic semantic division between proximal demonstratives (Sets A–C) versus distal demonstratives (Sets D–F). Pairs of these sets (e.g., Sets A/D, B/E, C/F) use the same demonstrative root, but the proximal and distal pairs use different inflectional endings. The roots and their approximate glosses are given in (4), abstracting away from inflection. Set C, the proximal pair to Set F, exists in some daughter languages but is not reconstructed to Proto-Algonquian. For Sets A/D, the root is *iw- when followed by animate singular inflection, but *iyo·- for all other persons and numbers.

(4) Proto Algonquian demonstrative roots (based on Goddard 2003: 56)

Sets A/D	'this'	iw-/iyo·-
Sets B/E	'that'	in-
Set F	'yonder' (rem.)	iy-

The reconstructions from Proulx (1988) are given below in Table 6. He posits a slightly different set of semantic distinctions than Goddard (2003), with five degrees of deictic difference (as in modern Blackfoot). The roots divide into a glide series and a nasal series, each of which combines with a different series of inflectional suffixes. Proto-Algonquian *w regularly deletes before a rounded vowel, so the proximal restricted root *o·- is part of the glide series. Note that both Proulx (1988) and Goddard (2003) largely agree on the first consonants in demonstrative roots (glides and nasals), even if they disagree on their distribution.

Table 6: Proto Algonquian demonstrative roots (based on Proulx 1988: 315)

	Proximal		Distal		
	restricted	extended	restricted	extended	remote
Glide	*o·- (< **wo·-)	*w-	*yo·-	*y-	*ye·-
Nasal	*mo·-	*m-	*no·-	*n-	*ne·-

Relevant for this paper, Goddard (2003) includes an initial *i in his reconstructions while Proulx (1988) does not. Both authors point out than many of the daughter languages have doublets with and without an initial vowel, or synchronic alternations where the initial vowel alternates with zero. Proulx reconstructs the vowel only where the daughter languages agree, and suggests that the initial vowel could have been a post-Proto-Algonquian addition because some demonstratives are otherwise sub-minimal words (Proulx 1988: 311, note 2). Goddard explicitly interprets the alternations with zero to be phonetic reduction or deletion as an external sandhi process when the demonstrative is non-initial in a phrase (Goddard 2003: 83, 86–87). Goddard (2018) adopts the reconstructions from Goddard (2003), but it is worth pointing out that an alternative reconstruction is available where Proto-Algonquian demonstrative roots had no initial *i at all.

Relative roots or stems license and require an oblique nominal element in the clause (Bloomfield 1946: 120, Rhodes 2010). These oblique nominals are distinct from other applicative arguments or locative-marked nominals (see discussion in Dahlstrom 2014, Lockwood & Macaulay 2024, Rhodes 2010, Xu 2022). Each relative root specifies a relationship between the verb and the oblique nominal it licenses, "which may serve to indicate the predicate's source, reason, manner, location, quantity," etc. (Valentine 2001: 421). For examples of relative roots in specific languages, see Costa 2017: 363–367 (Miami-Illinois), Dahlstrom 2014: 57–60 (Meskwaki), Kim 2020, 2023 (Blackfoot), Valentine 2001: 421 (Ojibwe), Wolfart 1973: 66 (Plains Cree). Like other roots, relative roots may occur as the initial element in a stem or as a derived preverb or particle. Some verb stems license oblique nominals even without a relative root.

The relative roots and stems in Proto-Algonquian which begin in *i are listed in (7). Glosses follow Goddard (2018: 99), where the curly braces indicate the general type of semantic predicate they license, like a variable that is specified in context by the oblique nominal itself or which else is deictic (Goddard 2002: 49, Goddard 2021: 44, Goddard & Thomason 2014: 8). I have included reconstructions from Goddard (amalgamated from Goddard 2002: 58, 60, Goddard 2018: 99) and Pentland (2023) (with pages given in the format [P #]). Goddard often reconstructs an abstract stem but also gives examples of reconstructed full words; Pentland reconstructs full words for inflectable stems. I have add surface morpheme breaks to full word reconstructions and replaced initial *e with *i where needed. The root *iθ- '{so}; to {somewhere}' regularly becomes *iš- before *i, including the final *i of a preverb or particle, as in PT *iši. The stem reconstructions require some interpretation, below.

Table 7: Proto-Algonquian relative roots and stems with initial *i

	Goddard (2002: 58, 60, 2018: 99)			Pentland (2023)	
	stem	*word*		*word*	
RT '{so}; to {somewhere}'	*iθ- ∼ *iš-	PT	*iši	*iθ-	[P 1332]
AI 'go to {somewhere}'	*iha·-	3s	*ihe·-wa	*ye·-wa	[P 3639]
AI 'say {so}'	*i- ∼ *iyo-	3s	*i-wa	*i-wa	[P 1396]
TA 'say {so} to'	*iθ- ∼ *Ø-	3→3′	*iθ-e·-wa,	*iθ-e·-wa	[P 1376]
		3′→3	*Ø-ik-wa		
TI 'say {so} to'	*itam-	3s	*itam-wa	*itam-wa	[P 1331]

For the AI stem 'go to {somewhere}', the stem ends in an underlying *a·, which ablauts regularly to *e· before the *w in the 3s suffix (Goddard 2007: 209). Goddard reconstructs this form with an initial *i and a following *h; Pentland reconstructs this form without an initial *i and with a semivowel *y. Meskwaki (Fox; listed under F in Pentland 2023) is the only language listed in Pentland (2023: 3639) whose 3s form begins with *i*, *ihe·-wa*. The 3s in all of the other languages begins in a vowel (e.g, Miami-Illinois (MiIl) *e·-wa, Massachusett (Ms) *a·-w, Munsee Delaware-Wappinger (Mu) *é·-w, and Shawnee (Sh) *he·-wa, where the initial *h* is epenthetic). Evidence for reconstructing an initial *i comes primarily from changed clauses, where "changed" or "initial change" in Algonquian refers to a system of morphological ablaut which affects the initial syllable of a word. The participle in Meskwaki is *ye·ya·-t-a*, where the initial *i has ablauted to *ye·. The changed form in other languages looks like an initial sequence *(y)é· is prefixed to the verb (e.g., the Massachusett participle *é·ya·-t*, or Shawnee changed conjunct *ye·ya·-č-i*). If this is understood as ablaut of an initial vowel, then there must be an underlying initial vowel *i which is deleted initially in all languages except Meskwaki.

The AI stem 'say {so}' is typically reconstructed as simply *i-; Goddard presumably lists the allomorph *iyo- because the 3p form in Meskwaki is *iyo·-w-aki* (Pentland 2023: 1396). Pentland reconstructs the participle form as *ye·-t-a, which is expected if the initial *i ablauts to *ye·. Additionally, this *i causes a preceding *t to assibilate to *s, hence 1s *nes-i 'I say {so}' (from *ne- '1s', epenthetic *t, and the stem *i-).[5]

The TA stem 'say {so} to' was originally analyzed as containing the relative root *iθ- ∼ *iš- '{so}; to {somewhere}' (Bloomfield 1925: 133, 145). Later, the TA and TI stems were reanalyzed as containing *i- AI 'say {so}' followed by the common transitivizing suffixes, TA -*θ* and TI *-tam (Bloomfield 1946: 99, 112, 120). The TA stem has a null allomorph *Ø- before

[5]Bloomfield (1946: 98) described this verb as "entirely irregular", but Pentland (1979: 403) points out that this is expected if Proto-Algonquian neutralized a contrast between short *i and *e in the initial syllable of the word. The verb 'say {so}' begins in a reflex pre-Proto-Algonquian **i, whereas stems with initial *i that do not cause assibilation begin in a reflex of pre-Proto-Algonquian **e. There are a small number of other forms whose internal reconstruction point to pre-Proto-Algonquian **i; see Pentland (1979: 403) and Pentland (2023: 1418).

inverse *-ekw and 2nd object *-eθ (Bloomfield 1946: 99, Pentland 2023: 1376). Presumably, a *t preceding either the TA or TI stems was assibilated to *s, just as for *i- AI 'say {so}', but all languages have restored *t except for Eastern Abenaki for the TI stem inflection.

Besides *iθ- ∼ *iš- '{so}; to {somewhere}', the other relative roots listed in Dahlstrom (2014), Lockwood & Macaulay (2024), Rhodes (2010) either begin in a short vowel *a or *o, or they begin in *t (initially) ∼ *-ent (non-initially); I return to these cases in Section 6. I do not know the full list of relative stems in Proto-Algonquian, but the few listed for Meskwaki in Dahlstrom (2014: 60) either begin in a short vowel or an obstruent. Thus, the relative roots and stems with initial *i are either followed by a sonorant (*h ∼ *y), or by *θ, or by nothing (as in *i- AI 'say {so}', and the derived TA and TI stems).

Due to these restrictions, initial short *i in Proto-Algonquian only occurred before sonorants in phonologically small roots and stems (e.g., demonstratives and relative roots), or else before *r, *θ,[6] or the cluster *šp and any cluster of the shape *Ck, where C is anything *except* a nasal or *s. This is an incredibly narrow phonological distribution. As a result, no roots in "Core Algonquian" languages begin with reflexes of Proto-Algonquian *ip-, *it-, *ič-, *ik-, *is-, *iš-, and no long stems begin in reflexes of *im-, *in-, *iw, or *iy. In the next section I turn to the distribution of initial *i* in Blackfoot.

5.2 Blackfoot data

Unlike "Core Algonquian" languages, many roots in Blackfoot apparently begin with an *i* or exhibit an [i] ∼ Ø alternation at their left edge. Goddard (2018: 99) points out that the dictionary, Frantz & Russell (2017), lists many Blackfoot verbs under *ip-*, *ik-*, etc, but very few under obstruents, except for *s-* . This can be confirmed by counting the number of entries under each section of the dictionary. An R script was used to extract entries from the original Word document underlying the dictionary.[7] These were placed into a text document with tab separated values which was then double-checked for accuracy and counted. The corpus contains 5,180 entries in total, including some entries which appear in the Word document but not in the published dictionary. I grouped the entry headers by morphological type and subclassification into one of four simplified categories: "noun", "verb", "adjunct", and "other", with details of this mapping described in AUTHOR (2026a). The "noun" and "verb" categories are noun stems and verb stems, respectively. The "adjunct" category follows the terminology from Frantz (2017), Frantz & Russell (2017), and corresponds to Bloomfield's preverbs and general roots (Bloomfield 1927, 1946). Crucially, "adjuncts" are prefixes and may occur in word-initial position, just like verbs and nouns. "Other" is a catch-all category for the remaining entries, including bound morphemes, and minor classes like demonstratives and uninflected stems. The dictionary headwords are alphabetized into 13 sections: a, h, i, k, m, n, o, p, s, t, w, y, '. The vowel sections contain short and long vowels, so I have further divided those sections by phoneme.

The counts of entries by phoneme and simplified category are shown in Table 8. Of note is the relative rarity of verb entries beginning in the plosives *p*, *t*, or *k*, or the nasals *m* and *n* (boxed below). In contrast, there is a much higher number of entries under *i* than the other two short vowels (boxed below).[8] The eight words listed under <h> are all uninflected interjections, a

[6] As discussed above, both of these sounds may have been phonetic laterals—a small and easily definable class.
[7] Thank you to REDACTED for writing this script.
[8] There are other notable asymmetries in these counts as well. The 29 entries listed under <t> are substantially

category for which *h* is predictable before vowel-initial words (Taylor 1969: 30). Outside of this category, Blackfoot words never begin in initial *h*. The count for <s> is higher than the other consonants because of the unusually wide distribution of *s* (see Goad & Shimada 2014a,b, Weber & Derrick 2025): this entry contains stems beginning with short *s* and long *ss* before vowels and consonants. Consequently, the count is roughly 4x larger than the other consonants.

Table 8: Number of dictionary headwords listed by section and simplified category

Section		**verb**	**noun**	**adjunct**	**other**	**Grand Total**
<p>		$\boxed{3}$	122	16	1	142
<t>		$\boxed{1}$	11	1	16	29
<k>		$\boxed{7}$	164	22	18	211
<m>		$\boxed{18}$	213	33	3	267
<n>		$\boxed{20}$	107	28	12	167
<s>		541	187	74	14	816
<w>		274	1	37	2	314
<y>		173	4	16	8	201
<h>		0	0	0	8	8
<a>	/a/	55	158	33	21	267
<i>	/i/	$\boxed{926}$	235	163	37	1361
<o>	/o/	569	188	74	19	850
<o>	/oi/	1	0	0	0	1
<a>	/aa/	19	92	15	3	129
<a>	/ai/	7	66	4	1	78
<a>	/ao/	1	38	1	1	41
<i>	/ii/	31	178	24	2	235
<o>	/oo/	22	34	6	2	64
Grand Total		**2667**	**1798**	**547**	**168**	**5180**

Moreover, many of the Blackfoot forms listed under <i> are cognate to "Core Algonquian" forms beginning in a consonant. Goddard (2018: 100–101) discusses six Blackfoot stems in total, along with their Proto-Algonquian (PA) reconstruction. The reconstructions and the Blackfoot entry header in Frantz & Russell (2017) are given in (5) below. As shown here, Blackfoot stems beginning in stops are listed under <i> (as in the first three rows), but stems beginning in *s* or *ss* are listed under <s> (as in the last three rows).

lower than the number of entries listed under the other consonants, which typically range between 150 and 250. There are a number of historical changes which have merged initial *t with other consonants in Blackfoot (Berman 2006: 265, 267, AUTHOR 2026b). There are also many verbs listed under <w> but few nouns, and there are many nouns listed under <aa> but few verbs. AUTHOR (2026a) argues that both groups begin with a phonemic long vowel.

(5)

Proto-Algonquian	Blackfoot	
*po·n- RT 'cease'	*ipon-*	vrt; terminate, end, be rid of [FR 91]
*ki·šow- RT 'warm', *-es TI1 'by heat'	*iksistohsi-*	vti; warm over a heat source [FR 63]
*ketem- (initial)	*iḳimm-*	vta; show kindness to, bestow power upon, care for [FR 46]
*keskinaw- TA 'know'	*ssksino-*	vta; know/think about [FR 269]
*keskin- TI1 'know'	*ssksini-*	vti; know [FR 268]
*sakipw- TA 'bite'	*siksip-*	vta; bite [FR 251]

This data suggests that PA verb stems beginning in *C- correspond to Blackfoot stems beginning in *iC-. Some of the stems in (5) do truly begin in *iC-* across all morphophonological contexts. One such stem is *-iḳimm-* vta; 'show kindness to, bestow power on, care for', underlined in (6), which begins in *i* word-initially in imperative clauses (6a), word-initially in indicative clauses (6a), and after a prefix (6a).

(6)
a. **i**kímmisa! 'bestow power on him!/care for her!' [FR 46]
b. **i**kímmiiwa 'he bestowed power on him' [FR 46]
c. nits**í**kimmoka 'he bestowed power on me' [FR 46]

But many of the other stems exhibit an initial *i(i)* ∼ Ø alternation. For example, the root *-ipon-* vrt; 'terminate, end, be rid of' begins in *p* word-initially in imperative clauses (7a), with a long *ii* word-initially in indicative clauses (7a), and either with *i* or *ii* after a prefix (7a).

(7)
a. ponihtáát 'pay!' [FR 91]
b. **ii**pónihtaawa 'he paid' [FR 91]
c. nits**i**pónihta/nits**íí**pónihtaa 'I paid' [FR 91]

Different stems apparently have initial *i* across different morphophonological contexts. For example, the stem *-siksip-* vta; 'bite' begins in *s* word-initially in imperative clauses (8a), with or without a preceding *i* word-initially in indicative clauses (8a), and with no *i* at all (but a lengthened initial *ss*) after a prefix (8a).

(8)
a. siksipísa 'bite thou him!' (Taylor 1969: 76)
b. siiksipáwa/**i**siksipáwa 'he was bitten' (Frantz 1971: 12)
c. nítssiksipawa 'I bit him' [FR 251]

Despite this variation, Goddard implicitly assumes that the Blackfoot forms beginning in a consonant are either more recent innovations or can be explained as regularities with a synchronic analysis. Goddard (2018: 99) argues that the morphophonology of person prefixes supports his hypothesis that Blackfoot stems begin in *iC-. In "Core Algonquian" the person prefixes are *ne(t)- '1', *ke(1)- '2', and *we(t)- '3'. The final *t only occurs before vowels, as in Menominee (M) *net-ākīhcem-a-w* TA 'I put him, her, it (animate) in or on water; soaks, floats, dips him, her, it (animate)' (Bloomfield 1975: 14). The final *t is absent before consonants, as in M *ne-pūnehsemi-m* AI 'I stop dancing' (containing *po·n- RT 'cease') (Bloomfield 1975: 224) or M *ne-sākep-ok* TA 'he or she bites me' (containing stem *sakipw- TA 'bite') (Bloomfield 1975: 229). Blackfoot person prefixes end with *t* (or *ts* before *i*) before all verb stems, as in *nits-ipónihtaa*

AI 'I paid' [FR 91] (containing *po·n- RT 'cease'), supporting the idea that they begin in a vowel. This is true even when the person prefix is followed by *ss* or *sss* instead of a vowel on the surface, as in ***nít**-ssiksip-a-wa* TA 'I bit him' (Frantz 2017: 176). Goddard (2018: 99) argues that the Blackfoot verb stems listed under <s> and <ss> begin abstractly with an initial /i/ in order to explain the allomorphy of the person prefixes.

Goddard interprets these facts to mean that "all Blackfoot verb stems basically begin with a vowel" (Goddard 2018: 99). Although he recognizes that many Blackfoot nouns are listed under a consonant in the dictionary, he does not take this to be the main pattern. He also implicitly assumes that the initial $i \sim Ø$ alternation for some roots is tractable, occurring "under synchronic conditions that have not been described". I take up a synchronic phonological analysis in Section 7.

5.3 The argument

The empirical fact to be explained is that Blackfoot apparently has many roots and stems with initial *i*, but the other Algonquian languages (referred to as "Core Algonquian", following Goddard 2018) only have initial *i* or *e* (< PA *i) in a small number of roots with a narrow phonological distribution. Goddard (2018) argues that Proto-Algonquian-Blackfoot (PAB) contained roots beginning with *iC-, but Proto-Algonquian (the ancestor of the "Core Algonquian" languages) underwent an innovation where initial *i- was lost by the rule in (9), leaving only a handful of remnants. This innovation (boxed below) defines a Proto-Algonquian subgroup, as in (10). Under this analysis, Blackfoot retains an archaic feature of Proto-Algonquian-Blackfoot, at least in verbs. The analysis does not account for why Blackfoot nouns frequently begin in consonants (see Table 8), nor does it account for the Blackfoot roots which exhibit initial $i \sim Ø$ alternations. I return to these questions in Section 7.

(9) Core Algonquian #i-loss
*i > Ø / #__C

(10) Proto-Algonquian-Blackfoot

*iC-

Blackfoot — Proto-Algonquian

*iC- — $\boxed{\text{*C-}}$
*iC- (remnants)

"Core Algonquian"

If *all* of the *C-initial stems in Core Algonquian with Blackfoot reflexes in *iC-* were derived from Proto-Algonquian-Blackfoot (PAB) *iC-, then PAB would have a typologically unusual lexicon, which few or no stems in beginning in *C-. Foreseeing this issue, Goddard (2018: 100) suggests that PAB very likely included roots beginning in *C- as well as *iC-, leading to the two correspondence sets in (11). This would mean that all of the daughter languages fully neutralized a contrast from PAB but in different ways: Proto-Algonquian lost initial *i by systematic sound change, and Blackfoot extended the *iC- pattern to the *C- roots by analogy. Without independent evidence to determine the original shape of the root, such a division is completely arbitrary and it would be simpler to either reconstruct *C- or *iC- for all roots. Here, the best reconstruction would be *C-, because it is the form that occurs in the majority of daughter languages, and it is the form that would create a typologically unmarked lexicon.

(11)

PAB	PA	Blackfoot
*iC-	*C-	iC-
*C-	*C-	iC-

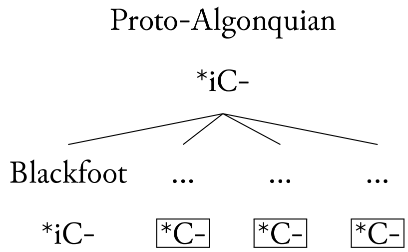
Goddard (2018: 101) suggests that archaic patterns of initial change provide independent evidence for the correspondences in (11). I do not find this argument convincing, and it is worth briefly explaining why. "Initial change" refers to a family-wide, reconstructable system of morphophonological processes which affect the first syllable of the word under particular morphosyntactic conditions (Costa 1996). For example, Goddard (2018: 100) discusses how the present of initial change (bolded) on the vowel *after s* in *s**ii**ksip-á-wa* TA 'he was bitten' (Frantz 1971: 12) means that this root began at one point in *s-, even though synchronically it begins with *is*- in some contexts. He concludes that some roots in PAB must have begun with *C-. There are two problems. First, *every* example of archaic initial change in Taylor (1967) supports a reconstruction beginning in *C-. With no independent evidence for roots beginning in *iC-, it is simpler to reconstruct all PAB roots as *C-. Second, the novel form of initial change simply prefixes *i*- or *ii*- (itself possibly a changed form of the *i*- prefix). This means that whenever we do encounter initial change on an initial *i*-, it is very likely a novel form of initial change rather than an archaic form. Possibly, future research into Blackfoot initial change will reveal more conclusive evidence for Goddard's proposal, but for now, I set aside initial change as a diagnostic for initial syllables. I return to patterns of initial change in Blackfoot in Section 7.2.2.

5.4 Summary and alternative analyses

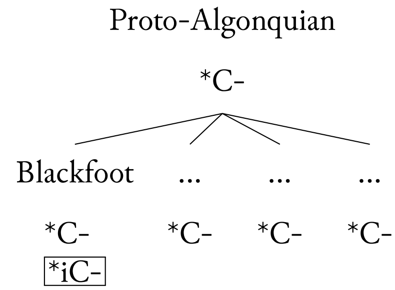
In summary, Goddard (2018) argues that Proto-Algonquian, but not Blackfoot, deleted stem-initial *i-. There are at least two plausible alternative explanations for why "Core Algonquian" languages share a feature that Blackfoot does not. Considering the difficulties of subgrouping discussed in Section 2, these alternatives must be refuted before we can conclude with Goddard that the "Core Algonquian" languages share an innovation, as in (10). First, even if the proto-language contained roots with initial *i (retained in Blackfoot), the "Core Algonquian" languages may have each lost initial *i independently due to parallel development or drift. This is represented in (12a), with a box around each individual innovation. Second, the proto-language may have contained roots with initial consonants (retained in "Core Algonquian" languages), and Blackfoot alone has innovated initial *i* in one or more phonological contexts. This is represented in (12b), with the Blackfoot innovation(s) boxed.

(12) Alternative explanations for shared features in “Core Algonquian”

a. Parallel development



b. Retention



I examine each of these alternative hypotheses in the next two sections. I conclude in Section 6 that there is no strong evidence against the parallel development hypothesis represented by (12a). In Section 7 I argue in favor of the hypothesis represented by (12b), and show that initial *i* in Blackfoot is best explained as several innovations in Blackfoot alone. The shared features in “Core Algonquian” are instead retentions from an ancestor language, with no strong evidence that they form a subgroup.

6 Confound: no strong evidence against parallel development

Sound changes are only good evidence for a shared innovation if the changes are unlikely to arise independently in each language. The loss of initial *i in “Core Algonquian” does not meet this bar. I argue in Section 6.1 that loss of initial *i is typologically common and especially expected under the conditions of Goddard’s proposal. In Section 6.2 I argue that the loss of *i also has family-specific phonetic motivations, which means that the sound change could easily have developed independently in each of the “Core Algonquian” languages.

6.1 Apocope is not an unusual sound change

Sound changes which are typologically common are not typically seen as good evidence for a shared innovation. Apocope (loss of an initial vowel) is explicitly noted by Clackson (2022: 25) as a frequent sound change which is non diagnostic of an innovation. In this case apocope would be well-motivated by syllable structure. As discussed above, the lexicon for PAB would contain many roots beginning in a vowel, and few or none beginning in a consonant. The optimal and unmarked syllable structure across all languages begins in a consonant, so apocope would simply be restructuring the lexicon towards an unmarked structure.

A shared sporadic sound change might be better evidence for an innovation, because it is unlikely that a sound change would affect the same set of random lexical items multiple times independently (Crowley 1997: 169). If individual languages lost initial *i independently at different times, then the remnant stems beginning in *iC- would likely be different for each language. It is true that all “Core Algonquian” languages have initial *i in the same set of roots. However, this set of roots is not randomly distributed throughout the lexicon. The roots share the same phonological restrictions in terms of size (e.g., demonstratives and relative roots/stems) and following consonant(s) (e.g., *i only occurs before sonorants in small roots, or before *r, *θ, or a small set of clusters in longer roots). The loss of *i is therefore not sporadic. Furthermore, there is a plausible alternative that *i was epenthesized in this set of roots in Proto-Algonquian (or independently in the daughter languages), as suggested by Proulx (1988) specifically for demonstrative roots (discussed in Section 5.1).

Even if apocope is a typologically common sound change, it might be taken as supporting evidence for a subgroup if the subgroup also aligns with many other innovations. However, the loss of initial *i is only one of two innovations proposed for "Core Algonquian" languages, and it is the only proposed *phonological* innovation. Without corroborating evidence from other uncontroversial innovations, I conclude that apocope is not strong evidence for a shared innovation or subgroup.

6.2 Phonetic motivation for initial *i loss

Goddard (2018: 100) posits that initial syllables in Proto-Algonquian were prosodically weak, and that the loss of initial *i was just one of several types of phonetic reduction in weak initial syllables. He points out two phonotactic constraints on initial syllables discussed in Section 4.1: initial syllables lack clusters of true consonants and lack a contrast between short *i and *e. Additionally, the allomorphy of relative roots beginning with PA *t- (initially) and *-ent- (non-initially) could also be explained by reduction in initial syllables. For example, the Proto-Algonquian relative root *taθ- '{somewhere}' has a non-initial allomorph *-entaθ- after a prefix, and an initial changed form *e·ntaθ-, (13). The Unami Delaware examples contain reflexes of this relative root (underlined) and are repeated from Goddard (2018: 100).

(13)		pre-PA	PA	Unami Delaware	Translation
	Initial	*entaθ-	*taθ-	té·kəne talá·wsu	'he lives in the woods'
	Non-initial	*entaθ-	*entaθ-	ntəntalá·wsí·ne·n	'for us to live there'
	Changed	*e·ntaθ-	*e·ntaθ-	yú entala·wsíenk	'here where we live'

An internal reconstruction of this pattern of allomorphy suggests that the relative roots in pre-PA began with something like the non-initial allomorph. For the example here, that would be pre-PA *entaθ- '{somewhere}'. The initial allomorph was reduced to *t in prosodically weak initial syllables, but the unreduced allomorph was maintained in non-initial syllables or when the vowel was lengthened to *e· under initial change.

The problem is that if the loss of initial *i in Proto-Algonquian is motivated by phonetic factors, then there is less reason to believe this was a shared innovation. Each daughter language would inherit prosodically weak initial syllables, and so initial *i could have plausibly been lost in each language independently due to the inherited phonetic factors. The short vowel *i would be the least sonorous of the short vowels, and therefore the most likely to undergo apocope.

In an alternative interpretation, perhaps the innovation in Proto-Algonquian is a prosodically weak initial syllable itself. Even if individual "Core Algonquian" languages loss initial *i at different times, the loss would follow from a shared feature that Blackfoot does not share. However, Blackfoot has the same restrictions on initial syllables as "Core Algonquian" languages. Just as in PA, no Blackfoot words begin in a true consonant cluster (*'C*, *hC*, or *ssC*), and there is no contrast between PA *i and *e in the initial syllable. In fact, *i and *e have merged to Blackfoot *i* everywhere, though only *i* < PA *i causes a preceding /k/ to assibilate to *ks* and a following /t/ to pre-assibilate to *st*; Berman 2006: 265.

Blackfoot also shows reduction in initial syllables without initial change. I know of no cognates in Blackfoot to the Proto-Algonquian relative roots that begin in *(en)t-. The process of reduction in Blackfoot is more general than this, and it affects other (relative and lexical) roots. The examples in (14) involve the relative root *oht-* 'instrumental, means, source, content, path' (Frantz 2017: 102) (< *went- RT 'from there, thence, therefore' Pentland 2023: 3415). This root is spelled by Taylor (1969) as *uxt-* (or *uxts-* before *i*). As shown here, this relative root is pronounced as *t-* in initial position, but *uxt-* in non-initial position. The changed form is *ixt-*, involving an ablaut of *u* to *i*.

(14)	Initial	tsítskixpissi	'when he danced by'	[T 134]
	Non-initial	áxkuxtsitokoopsskaaʔwa	'so she can make broth with (them)'	[T 304]
	Changed	ixtsítsksssapiʔwa[9]	'he is looking past'	[T 63]

An internal reconstruction suggests that this relative root in pre-Blackfoot was *oht- in initial and non-initial positions, and that the initial allomorph was reduced to *t-. The reconstructions below use the orthography from Frantz (1978, 2017) and abstract away from the regular process of *t assibilation before *i, *i·, or *j.

(15)		pre-Bl		Blackfoot
	Initial	*oht-	>	ts-
	Non-initial	*oht-	>	uxts-
	Changed	*iiht-	>	ixts-

The same process of reduction holds in other roots as well. This is shown in (16) for the relative root *p-* ∼ *ohp-* 'associative' (Frantz 2017: 102) and in (17) for the root *k-* ∼ *ohkan-* 'all'.

(16)	Initial	pópaatsisa!	'hold him on your lap!'	[FR 176]
	Non-initial	áakohpópaatsiiwáyi	'she will hold him on her lap'	[FR 176]
	Changed	iihpópaatsiiwa	'he held [the baby] on his lap'	[FR 176]

(17)	Initial	kaná'pssik!	'you (pl.) gather for a rodeo!'	[FR 164]
	Non-initial	áakohkaná'pssiyaawa	'they will gather for a rodeo'	[FR 164]
	Changed	iihkaná'pssiyaawa	'they gathered for a rodeo'	[FR 164]

Blackfoot exhibits the same phonotactic constraints and phonetic reduction in the initial syllable as the "Core Algonquian" languages. In other words, the prosodic weakness of the initial syllable is not a shared feature of "Core Algonquian" to the exclusion of Blackfoot. If prosodic weakness motivated the loss of initial *i in the "Core Algonquian" languages, then the proposal in Goddard (2018) cannot explain why Blackfoot failed to undergo *i- loss. I conclude there is no strong evidence against parallel development, where individual languages independently have lost initial *i and Blackfoot happens to be the one language that has not.

Instead, I consider an alternative analysis where the "Core Algonquian" languages do not form a subgroup at all and instead retain initial *C- from a shared ancestor. Under this analysis, Blackfoot has innovated stems with initial *i under certain conditions. Arguing for this requires a full synchronic analysis of root alternations in Blackfoot, which I take up in the next section.

[9] The original transcription was <ixtsítsksspaiʔwa>, with the <p> and <a> metathesized from correct *ixtsítskssapiʔwa*. The corrected form contains the stem *ssapi-* AI 's.o. looks' which fits with Taylor's translation.

7 Alternative: retention in "Core Algonquian" and innovation in Blackfoot

I argue that the stems beginning in *C- in "Core Algonquian" are shared retentions from a common ancestor. In Section 7.1 after a brief note about how to interpret the dictionary entries in Frantz & Russell (2017), I point out that there are many Blackfoot roots which do begin in *C-* in initial positions. These forms I interpret as the regular retention from a common ancestor. The forms beginning in *i* are better explained as an innovation in Blackfoot. In Section 7.2 I take up a synchronic analysis of root alternations and argue that initial *i* has been added to Blackfoot roots in two different positions. Some roots have extended the *i* to all positions, lexicalizing the *i* as part of the root. In Section 7.3 I show that this can be confirmed by older documentation of these forms in the historical record. Section 7.4 concludes.

7.1 Blackfoot shares initial *i-loss

Importantly, the headwords in the dictionary do not always reflect a word-initial variant of the root or stem. (It is unclear to me if Goddard realized this or not.) As Frantz and Russell (2017:*xx–xxi*) explains, the headwords for non-stative verbs come from the form of the verb in a non-initial position (typically after the durative prefix *á-*), with the prefixes and any inflectional suffixes removed. The headwords for all other types of stems, including stative verbs and nouns, come from unprefixed stems with inflectional suffixes removed. Consequently, non-stative verbs are listed under their non-initial variants, while nouns and stative verbs are listed under their initial variants. Crucially, all of the examples from Goddard (2018: 99–100) listed in (5) are non-stative verbs and the headword therefore reflects a non-initial variant. Potentially, the *initial* variant of these stems begins in a consonant, while the non-initial variant (reflected in the headword) does not.[10] Generalizing beyond the six forms in (5), the gaps in Table 8 could be partially explained if *non-initial variants* of verbs never begin in a stop.

Blackfoot nouns and stative verbs (e.g., the forms which *are* listed under consonants in the dictionary) are very often cognate to "Core Algonquian" forms beginning in a consonant. In Table 9 I give examples of Blackfoot nouns which are cognate to Proto-Algonquian forms beginning in an oral or nasal stop. These particular reconstructions are from Berman (2006), except I have replaced any Blackfoot roots or stems that he cites with full inflected nouns instead. All Blackfoot forms are from Frantz & Russell (2017) except for 'osprey', which is from Taylor (1960: 12). These Blackfoot nouns begin in *C-*, just like the other "Core Algonquian" languages.

This type of correspondence is not explained by the proposal in Goddard (2018). If PAB had stems beginning in *iC-, then Blackfoot must have a regular rule of initial *i-loss which is similar to the rule in "Core Algonquian", but which applies to nouns only, as in (18).

(18) Blackfoot #i-loss
$\quad$ $^{*}\mathrm{i} \rightarrow / \#\_\_\mathrm{C}$, in nouns only

In Table 10 I give examples of Blackfoot stative verbs which are cognate to Proto-Algonquian forms beginning in an oral or nasal stop. These particular reconstructions are from Berman (2006), except I have replaced any Blackfoot roots or stems that he cites with full inflected

[10] Note that Goddard's argument that Blackfoot stems begin in *i* because of the morphophonology of the person prefixes also only provides evidence that the *non-initial form* of stems begins in *i*.

Table 9: Correspondences between Blackfoot and Proto-Algonquian nouns

Blackfoot		Proto-Algonquian		
ksísskstakiwa	'beaver'	*ki·šk-	'cut, chop, sever'	Br 273
ksiistsikóyi	'day'	*ki·šekatwi	'day'	Br 273
kóóna	'frozen water, ice'	*ko·na	'snow'	Br 273
saahkómaapiwa	'boy'	*taxkw-	'short'[11]	Br 272
paahtsííksistsikomma	'osprey (lit. false thunder)'	*peʔt-	'by accident, mistake'	Br 273
pisstóóhtsi	'inside'	*pi·nt-	'inside'	Br 274
pomísi	'fat (dripping), lard'	*pemyi	'oil'	Br 280
maká'pssáísskiyi	'weed'	*mekw-	'trouble'	Br 268
moksísa	'awl'	*mekwi	'awl'	Br 278
matsówohkitópii	'rider on a fine horse'	*melw-	'good, well'	Br 275
miníyi	'island'	*men-	'mass, pile'	Br 275
mííni	'berry'	*mi·ni 'berry'	Br 273	

nouns instead. All Blackfoot forms are from Frantz & Russell (2017), unless otherwise marked. In all cases, the Blackfoot nouns begin in *C*-, just like the other "Core Algonquian" languages.

Table 10: Correspondences between Blackfoot and Proto-Algonquian statives

Blackfoot		Proto-Algonquian		
ksiistsikóʔwa	'it is day' [T 104]	*ki·šekatwi	'day'	Br 273
saahksstssímma	'he is light in weight'	*la·nk-	'light in weight'	Br 274
saaxkssíʔwa	'he is short' [T 77]	*taxkw-	'short'	Br 272
paahtsá'piiwa	'it was a mistake' [FR 79]	*peʔt-	'by accident, mistake'	Br 273
maká'piiwa	'it is bad'	*mekw-	'trouble'	Br 268
niipówa	'it was summer' [FR 313]	*ni·penwi	'it is summer'	Br 281

Below I have modified the rule in (18) so that it also applies to stative verbs.

(19) Blackfoot #i-loss (modified)
$*i \rightarrow / \#\_\_C$, in nouns and in stative verbs

Finally, some forms of non-stative verbs also frequently begin in a consonant even when they are listed under a vowel in the dictionary. In Table 11 I give examples of Blackfoot imperatives which are cognate to Proto-Algonquian forms beginning in an oral or nasal stop. These particular reconstructions are from Berman (2006), except I have replaced any Blackfoot roots or stems that he cites with full inflected imperatives instead.[12] All Blackfoot forms are from Frantz & Russell (2017) except for 'stop thou', which is from Taylor (1967: 154). In all cases, the Blackfoot imperatives begin in *C*-, just like the other "Core Algonquian" languages. Note

[12] Berman (2006: 271) actually said the Blackfoot stem for 'bite' is a reflex of the PA changed form *se·kipw-.

that only some of the examples that (Goddard 2018: 99–100) discusses are listed here (e.g., 'pay' and 'bite' from Section 5.2); other forms begin with an initial *i* in imperative clauses as well, which will be discussed in Section 7.2.3.

Table 11: Correspondences between Blackfoot and Proto-Algonquian imperatives

Blackfoot		Proto-Algonquian		
koonít!	'take it down (the tipi)!' [FR 56]	*kawen-	'push down'	Br 267
kaamó'sit	'steal!' [FR 43]	*kemot-	'steal'	Br 272
ksisskámmisa	'display dislike towards her!' [FR 66]	*ki·hka·-	'berate'	Br 274
ksimínihkatsisa!	'call him by his nickname!' [FR 60]	*ki·m-	'secretly'	Br 267
pisstááhkio'toosa!	press it in!' [FR 96]	*pi·nt-	'inside'	Br 274
ponihtáát	'pay!' [FR 91]	*po·n-	'cease, stop'	Br 267
siksipísa	'bite thou him! (Taylor 1969: 76)	*sakipw-	'bite'	Br 271
saaksít!	'go out!' [FR 238]	*sa·k-	'out'	Br 268
niipoyít!	'stop thou!'	*ni·pawi-	stand	Br 273

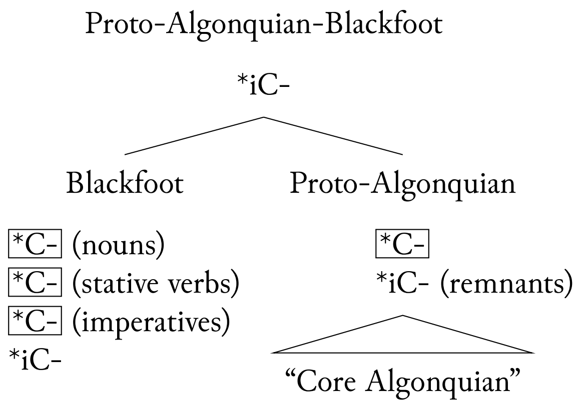
Below I have modified the rule in (19) so that it also applies to stative verbs.

(20) Blackfoot #i-loss (modified)
*i → / #___C, in nouns, stative verbs, and imperatives

The full set of morphosyntactic environments where stems surface with an initial *C*- is more complicated than this. Often, subjunctive clauses also begin with initial *C*-, but subjunctive clauses are not often found in the dictionary examples. Third person indicative clauses sometimes begin in a consonant (or optionally so), such as *ma'takit!* AI 'take (choose) (s.t.!)'. This seems to be changing over time, and Taylor (1967, 1969) includes more forms like this than Frantz & Russell (2017). From my own discussions with speakers, there is a lot of speaker and dialect variation in this initial position, which I return to in Sections 7.2.2 and 7.2.3.

The proposal in Goddard (2018) is that the loss of initial *i in the "Core Algonquian" languages is a shared innovation which defines a subgroup to the exclusion of Blackfoot. However, it turns out that Blackfoot shares this rule in several types of stems, including (at least) nouns and stative verbs. A revised picture of Goddard's proposal is given in (21) below. If PAB had only roots beginning in *iC-, then both Blackfoot and Proto-Algonquian independently lost initial *i in many of the same contexts (boxed; additionally, some of the non-stative verbs from Goddard 2018 have unexplained alternations with initial *C*-). This makes *i-loss look less like an innovation in Proto-Algonquian alone, and more like repeated parallel development–which I argued in the previous section is likely.

(21) Revised proposal (after Goddard 2018)



An alternative would be to treat the shared feature (initial *C- as a retention from the shared ancestor and to treat the unique feature in Blackfoot (initial *iC-) as an innovation. The next section presents a synchronic analysis which supports this alternative.

7.2 Synchronic morphophonological analysis

This section examines the left edges of roots in three different morphosyntactic contexts and show that initial *i* has been added in Blackfoot in different ways in these three contexts. Section 7.2.1 focuses on stems in non-initial positions. Root alternations reveal that all stems begin in underlying vowels and consonants in initial positions, but that this contrast is neutralized non-initial positions where all stems begin in a vowel. In non-initial positions, an *i is added to the beginning of stems that begin in an obstruent, which is the first case of innovated *i*. In Section 7.2.2 I turn to unprefixed non-stative verbs in indicative clauses. There is a novel form of initial change which is used in this position, and the form of initial change before obstruents is again the addition of *i* (or *ii*). Finally, Section 7.2.3 examines verbs in initial position in other clause types. For some verbs, the non-initial allomorphy of the stem has been extended to these positions as well and the verb has simply been reanalyzed as a vowel-initial stem. For stems that historically began in obstruents, this means that the *i* at the left edge of the non-initial variant is now used in all positions.

7.2.1 Non-initial variants

In this section I consider root alternations when they root stands in word-initial position versus after a prefix. These were recognized as important diagnostic positions in earlier descriptive work, which spent considerable space discussing these types of alternations (Taylor 1969, Uhlenbeck 1938). These two environments are also recognized as important in the Frantz & Russell (2017) dictionary, and many of the entries include examples with a stem or morpheme in both positions. To constrain the problem, I restrict myself to roots whose first vowel is short and followed by a single consonant – in other words, roots whose first syllable is light. Although there are many more patterns of alternation for roots which begin in a heavy syllable, this subset of the data is systematic and illustrates the fact that non-initial allomorphs always begin in a vowel in Blackfoot.

As Berman (2006: 267) and Goddard (2018: 99) noticed, stems which are non-initial in the word always begin with a vowel or glide (sometimes only abstractly). There is language internal evidence for this in root alternations. Below, I summarize a study that focuses on root alternations in verbs where the first syllable is light (e.g., V or CV). The relevant positions are at the left edge of the word (word-initial position) versus after a prefix (non-initial position). I will count left edge contexts as either an imperative, an unprefixed stative verb, or an unprefixed noun. As I discuss in the next sections, in many stems the non-initial form of the stem has been extended to word-initial positions, but these contexts are the most conservative. Roots fall into several patterns of alternation, depending on the segment at the left edge.

Any root that begins in an obstruent at the left edge, *C-*, has a non-initial variant in *iC-*. For example, the root *pon-* 'cease' begins in a *p-* at the left edge of the word, (22a), but begins in *ip-* after a prefix, regardless of whether the prefix ends in a consonant (22b) or a vowel (22c). I box the $\boxed{\text{i}}$ in the orthography line to highlight the additional vowel. The final vowel of the prefix in (22c) coalesces with the vowel *i* at the left edge of the root to form long *ai* [ɛː].

(22) a. Left edge

ponipisa!
[**p**on–p]–is–Ø
[cease–by.mouth.TA]–2s:3.IMP–CMD

'stop carrying him (e.g. a pup)!' [FR 91]

b. After C

áaks$\boxed{\text{i}}$***p****onipiiwáyi*
aak–[**p**on–p]–ii–Ø–w=áyi
FUT–[cease–by.mouth.TA]–3SUB–IND–3=PRX.PL

'she will stop carrying him with her teeth' [FR 91]

c. After V

áka$\boxed{\text{i}}$***p****onipawa*
akaa–[**p**on–p]–a–Ø–wa
PRF–[cease–by.mouth.TA]–3OBJ–IND–3

'he [pup] is no longer being carried by its mother' [FR 91]

This pattern holds true for stems with initial *p* and *k*. (As discussed in Section 5.2, there are very few stems beginning in *t* in Blackfoot.) For stems with initial *s*, there is no overt *i* after /t/ or /k/ and instead the /s/ is lengthened and fills the nucleus of the syllable. For example, the root *sim-* 'stab' begins in a *s-* at the left edge of the word, (23a), with *is-* after a consonant, (23b), but with *ss-* after a vowel (23c). I box the $\boxed{\text{i}}$ or $\boxed{\text{s}}$ in the orthography line to highlight the additional segment. The final vowel of the prefix in (23c) coalesces with the vowel *i* at the left edge of the root to form long *ai* [ɛː]. This is the pattern as well for the stem *siksip-* vta; 'bite' discussed by Goddard (2018).

(23) a. Left edge

simísa!
[**s**im]–:s
[stab.TA]–2s:3.IMP

'stab him!' [FR 251]

b. After C

*nít$\boxed{s}$**s**imoka*
nit–[**s**im]–ok–a
1–[stab.TA]–INV–3

'he stabbed me' [FR 251]

c. After V

*nitá$\boxed{í}$**s**imawa*
nit–a–[**s**im]–a–a
1–IPFV–[stab.TA]–3OBJ–3

'I am stabbing him' [FR 251]

Other roots that begin in an obstruent at the left edge of the word surface with an initial *oh* after a prefix, as shown in (24). The root [pomm-] 'buy' begins in a *p-* at the left edge of the word, (24a), but begins in *ohp-* after a prefix, regardless of whether the prefix ends in a consonant (24b) or a vowel (24c). The final vowel of the prefix in (**??**) coalesces with the vowel *o* at the left edge of the root to form long *ao* [ɔː]; after syllabification this vowel occurs in a closed syllable and surfaces as short. These roots can be analyzed as beginning in a consonant cluster, with deletion of *h* in initial positions and epenthesis of *o* in non-initial positions. I box the $\boxed{o}$ in the orthography line to highlight the additional vowel.

(24) a. Left edge

***p**ommáát!*
[**hp**omm–aa]–t
[buy–AI]–2S.IMP

'buy!' [FR 175]

b. After C

*áak$\boxed{o}$**hp**ommaawa*
aak–[**hp**omm–aa]–t
FUT–[buy–AI]–2S.IMP

'she will buy' [FR 175]

c. After V

*á$\boxed{o}$**hp**ommaawa*
a–[**hp**omm–aa]–wa
IPFV–[buy–AI]–3

'she is shopping' (BB)

Any root that begins in a nasal at the left edge, *N-*, has a non-initial variant in *Ø-*. For example, the root *misam-* 'long in time' begins in an *m* at the left edge of the word, (25a), but there is no nasal after a prefix, regardless of whether the prefix ends in a consonant (25b) or a vowel (25c).

Roots that begin in a nasal at the left edge of the word surface without a nasal after a prefix, as shown in (25).

For example, the root [misam-] 'long in time' begins in a *m-* at the left edge of the word, (25a), but no nasal after a prefix, regardless of whether the prefix ends in a consonant (**??**) or a vowel (**??**). I use a □ in the orthography line to indicate the missing nasal. The final vowel of the prefix in (**??**) coalesces with the vowel *i* at the left edge of the root to form long *ai* [ɛː].

(25) a. Left edge

***m**isámipaitapi'ssina!*
misám–[ipa–itapi]–'ssin–a
long.in.time–[life–person]–COLLEC–PRX

'people of long ago' [FR 150]

b. After C

*áaks□**i**samowa*
aak–[**m**isam–o]–w=áyi
FUT–[long.in.time–II]–IND–3=PRX.PL

‘it will be a long time’ [FR 97]

c. After V

*áka□**i**samowa*
akaa–[**m**isam–o]–wa
PRF–[long.in.time–II]–IND–3

‘it's been a long time’ [FR 145]

There are also roots that begin in a vowel in all positions. For example, the root *itsin-* ‘among’ begins in an *i* at the left edge of the word, (26), after a consonant (27), or after a vowel (28). The final vowel of the prefix in (28) coalesces with the vowel *i* at the left edge of the root to form long *ai* [ɛː].

(26) Left edge

***i**tsínohtoot*
[**i**tsin–oht]–oo–t–Ø
[among–put.TI]–TI2–2S.IMP–CMD

‘place it among the rest!’ [FR 120]

(27) After C

*áak**i**tsinohtoomáyi*
aak–[**i**tsin–oht]–oo–m–Ø=ayi
FUT–[among–put.TI]–TI2–IND–3=PRX.PL

‘he will place it among the rest’ [FR 120]

(28) After V

***áí**tsinohtoomáyi*
a–[**i**tsin–oht]–oo–m–Ø=ayi
IPFV–[among–put.TI]–TI2–IND–3=OBV.S

‘he is placing it among the rest’ [FR 120]

Those patterns are summarized in (29). As shown here, roots may begin with either a consonant or a vowel at the left edge, but all three patterns of roots begin in a vowel after a prefix. In other words, there is neutralization of contrast after a prefix (Weber 2021, 2023). Additionally, there are no roots which begin in a consonant in both positions.

(29)

Left edge	After C	After V	UR	Gloss
pon-	-ipon-	-ipon-	/pon-/	RT ‘cease’
sim-	-ssim-	-isim-	/sim-/	TA ‘stab’
pomm-	-ohpomm-	-ohpomm-	/hpomm-/	RT ‘buy’
misam-	-isam-	-isam-	/misam-/	RT ‘long in time’
itsin-	-itsin-	-itsin-	/itsin-/	RT ‘among’

These facts together mean that the best synchronic analysis is that roots which alternate are best analyzed as beginning in underlying consonants or consonant clusters. There are synchronic processes which epenthesize /i/ before obstruents after a prefix,[13] epenthesize *o* before /hC/ clusters after a prefix, and which delete a nasal after a prefix.

[13]The *i* at the left edge of *pon-* ‘cease’ has the same properties as the usual epenthetic [i] in Blackfoot; e.g. it causes a preceding /k/ to assibilate to [ks].

The *i* at the left edge of non-initial roots that begin in obstruents accounts for many of the cognates that Goddard (2018) noticed. Crucially, the synchronic morphophonological analysis above shows that the *i* must be added to an underlying form that begins in an obstruent. A synchronic analysis where vowels are deleted at the left edge would predict that we should also see alternations for vowel-initial roots like /misam-/ 'long in time', but we do not. Recall as well that many verbs are listed in the dictionary under their non-initial variant; these processes of epenthesis and deletion explain why so few verbs are listed under consonants. I now turn to two word-initial environments that also have added an initial *i*-.

7.2.2 Novel initial change

Novel forms of initial change in Blackfoot add a word-initial *i* in certain morphosyntactic contexts. "Initial change" in Algonquian is a set of morphophonological processes such as ablaut and *-ay- infixation which affect the first syllable of the word under certain syntactic conditions (Costa 1996). In Blackfoot the system is being restructured and simplified, so that initial change for consonant-initial words is now marked with a *i*- or *ii*- prefix. There are remnants of the archaic system which exist optionally alongside the novel forms for ∼100 stems (Taylor 1967).

Initial change in Blackfoot today is one of several possible ways to mark past tense on unprefixed verbs (Frantz 2017: 39–41), but it used to be used in a wider range of contexts as well (Taylor 1967, 1969: 120*ff*). The archaic patterns of initial change takes several forms: a short vowel becomes long *ii*, and a long vowel either changes vowel quality, infixes *ay* or *iy*, or a word-initial long *aa* becomes stressed (Frantz 2017: 39–41, Taylor 1967, 1969: 120*ff*, Weber 2020: Appendix C). These patterns are largely continued for verbs that begin in vowels, but a novel form of initial change is now being used for verbs which begin in consonants. The novel form of initial change simply prefixes an *i*- or *ii*- to the stem. The following examples illustrate archaic and novel initial change for three roots in Blackfoot. I have underlined the root and bolded initial change.

(30)	Initial change as vowel lengthening			
	Plain	siksip-ís	TA 'bite thou him!'	(Taylor 1969: 126)
	Changed (archaic)	s**ii**ksip-á-wa	TA 'he was bitten'	(Frantz 1971: 12)
	Changed (novel)	**i**siksip-á-wa	TA 'he was bitten'	(Frantz 1971: 12)
(31)	Initial change as vowel quality ablaut			
	Plain	poonixtáts-isa	TA 'pay thou him!'	(Taylor 1967: 154)
	Changed (archaic)	p**aa**nixtáts-isa	TA 'pay thou him!'	(Taylor 1967: 154)
	Changed (novel)	**ii**pónihtaa-wa	AI 'he paid'	[FR 91]
(32)	Initial change as -*ay*- infix			
	Plain	saaksí-t	AI 'go out!'	[FR 238]
	Changed (archaic)	s**ay**áaksìniisi-ʔwa	AI 'she had a miscarriage'	(Taylor 1969: 222)
	Changed (novel)	**íí**saaksini-wa	AI 'she miscarried'	[FR 233]

As shown in these examples, the novel forms of initial change add an *i*- to the left edge of stems in some word-initial contexts. Specifically, because initial change marks past tense in Blackfoot, this *i*- only occurs on non-stative verbs in realis/tensed clauses, or on states that have been coerced into a change of state reading. The novel form of initial change would not be found in imperatives or other non-finite clauses. However, some stems have been reanalyzed to have an *i-* in those contexts as well, to which I turn next.

7.2.3 Analogical leveling

Some Blackfoot verbs are regular reflexes of PA stems which begin in *C-, and yet synchronically begin in *iC-* in all positions. The reason is that the non-initial variant of the stem is now being used in all positions, which is leveling out the alternations. This affects word-initial position in morphosyntactic contexts that do not require initial change, such as the imperative for non-stative verbs, or the third person indicative for stative verbs. Recall that stems which begin in underlying obstruents have non-initial variants with an additional *i-* at the left edge of the stem, so these stems in particular begin in *iC-* while the cognates in other languages begin in *C-. One example of a stem which historically began in an obstruent and which now begins in *iC-* is *ik̲imm-* 'pity s.o.; care for s.o.' < PA *ketem-, repeated below from (6).

(33) a. **i**kímmisa! 'bestow power on him!/care for her!' [FR 46]
b. **i**kímmiiwa 'he bestowed power on him' [FR 46]
c. nits**í**kimmoka 'he bestowed power on me' [FR 46]

For other stems, there are two variants of the imperative listed in the dictionary (one in *C-* and one in *iC-*). One example is *-iksistohsi-* vti; 'warm over a heat source' < PA *ki·šow-es-. The regular reflex of this stem is the first listed imperative, *ksiistohsít!*, with a long vowel in the first syllable. Long vowels shorten in non-initial stems after the addition of *i-* (Berman 2006: 267–268), which is evident in the third example below. The second listed imperative levels out the root alternations by using the non-initial variant in this position as well, which contains the *i-* addition and also a short vowel in the following syllable.

(34) a. ksiistohsít!/**i**ksístohsit! 'warm it!' [FR 63]
b. **i**ksístohsima 'he warmed it, them (inan.)' [FR 63]
c. nits**í**ksistohsii'pa 'I warmed it' [FR 63]

It seems most likely that verbs like *ik̲imm-* in (33) have simply been reanalyzed as beginning in a vowel, and that verbs like *ksiistohsi-* in (34) are currently shifting in that direction. This is the second way in which an initial *i-* was innovated at the left edge of the stem in word-initial contexts.

7.3 Historical record

In many cases, it is possible to find older forms of words in the historical record which show that they began in consonants. To take an example like *ik̲imm-* vta; 'pity s.o.; care for s.o.', we can search for the imperative form across multiple sources. In all of the sources older than Frantz & Russell (2017), the imperative is recorded with an initial *k*. This shows that the paradigm was only leveled recently.

(35)	ikímmisa!	‘bestow power on him!/care for her!’	[FR 46]
	kímmisa!	pity thou him!	[T 79]
	kǐm’ǐs	pity him!	Tims (1889: 154)
	kǐm’okǐt	pity me!	Tims (1889: 154)
	kímmokit	pity me!	Uhlenbeck (1938: 232)

There are similar findings for stems which allow variation in word-initial position, such as -*iksistohsi*- vti; ‘warm over a heat source’. This is listed only with an initial *k* in older sources, such as in *ksǐstusit’* ‘warm it’ (Tims 1889: 186).

Furthermore, when there are changes to roots over time in the historical record of Blackfoot, the changes are always to *add* an initial *i* rather than delete one. All evidence shows that Blackfoot and “Core Algonquian” retain *C- initial roots from a shared ancestor, and that the only innovations are in Blackfoot root paradigms.

7.4 Summary

The evidence from synchronic root alternations and the historical record therefore converge on an analysis where Blackfoot historically added an *i* to the left edges of stems in several contexts. First, non-initial stems always begin in a vowel in Blackfoot. There is a conspiracy of processes which create this effect, and among those is the addition of /i/ to the left edge of stems which begin in an obstruent. Second, there is a novel form of initial change involving the prefixation of *i*- or *ii*- to words which begin with a consonant. Third, many stems have been relexicalized as their non-initial form, which means that vowel-initial allomorphs are now being used in other word-initial contexts as well.

As a result, the *i* at the left edges of Blackfoot stems results from innovations in Blackfoot alone, and many of these stems can be internally reconstructed as beginning in a consonant. This means that the initial consonants in Blackfoot as well as the other “Core Algonquian” languages are retentions from an ancestor language, and there is no evidence for a subgroup that excludes Blackfoot. This is schematized in (36), with boxes around the Blackfoot innovations.

(36) Proto-Algonquian

*C-

Blackfoot ... ...

*C- ~ $\boxed{\text{*-iC-}}$ *C- *C- *C-

$\boxed{\text{*iC-}}$ ~ $\boxed{\text{*-iC-}}$

This analysis allows us to reconstruct a more typical lexicon to Proto-Algonquian (e.g. C-initial verb stems). There is still a question about whether or not Proto-Algonquian contained any stems at all beginning with *i, or whether this was a true gap. As discussed in Section 5, many “Core Algonquian” languages have a small set of roots that begin in *i. These are traditionally reconstructed to Proto-Algonquian with an initial *i, but a plausible alternative is that these roots also began in consonants at some earlier stage (possibly pre-Proto-Algonquian) and that individual languages (or Proto-Algonquian) epenthesized an *i in these roots. Crucially, there is no evidence that this was an innovation of the "Core Algonquian" languages, because Blackfoot also contains a reflex of *i in cognate roots, such as *ist-* 'thus, there' from the relative root *iθ- '{so}; to {somewhere}'.

8 Discussion

The analysis in the previous section relied on a synchronic analysis of Blackfoot root alternations, as well as additional supporting evidence from the historical record. In this section, I discuss the importance of both of these methods, and end by raising the question of whether structural loss is good evidence for an innovation.

8.1 How synchronic analysis informs subgrouping

Hale (2015: 151–155) discusses a cases where evidence from allophony evidence affects subgrouping. In Blackfoot, the evidence from allomorphy is crucial evidence that affects subgrouping. The reason is that multiple hypotheses may be compatible with the data if we only look at a single morphophonological position. For example, in word-initial contexts it is true that Blackfoot stems frequently begin in *iC-* whereas their cognate forms in the other Algonquian languages begin with *C-*. Given this distribution, there are two equally good hypotheses about subgrouping: either Blackfoot retains an older pattern and the other languages underwent a shared innovation (as in Goddard 2018), or Blackfoot underwent innovations and the other Algonquian languages retain an older pattern (as argued here).

To distinguish between these two hypotheses, we must consider root alternations in addition to the static distribution of sounds across words. These morphophonological alternations allow us to posit underlying forms and an internal reconstruction of Blackfoot. There are a few takeaways from this work. First, stems should not be compared without their morphophonological and morphosyntactic contexts. For example, a word-initial *i* in Blackfoot is only surprising if it occurs in a morphosyntactic context where we would not expect initial change. Second, subgrouping arguments should be treated with caution unless they are supported by synchronic analyses.

8.2 A historical lexical database of Blackfoot

The reconstruction of the Algonquian family by Bloomfield (1925, 1946) is proof that the comparative method can be used without a historical record. That being said, when a language has undergone such stark and recent innovations as Blackfoot has, a historical record can confirm some of those sound changes. The Blackfoot Words database is a resource created in part to more easily find patterns in root alternations like the ones in this paper and to identify diachronic changes. This is a relational database of inflected words, stems, and morphemes in Blackfoot which have been digitized from sources representing all four major dialects and spanning the years 1743–2017. The most recent version of the database (v1.2; 2025-10-15) includes 4,557 word tokens from nine grammars, dictionaries, and wordlists. Each word token is fully analyzed into its constituent stems and morphemes, and tagged for morphosyntactic information. Each stem and morpheme token is linked to a standardized type called a "lemma", where the full list of lemmas functions more like a traditional wordlist or dictionary.

Neither morphophonological alternations nor diachronic change are directly encoded in the database. However, because each lemma links to all tokens of a stem or morpheme in the database, researchers may use a lemma to view the concordance of linked tokens. Relative to root alternations, the examples in the entries of Frantz & Russell (2017) do not always show a root in all three relevant positions (the left edge, after a consonant, and after a vowel). The database includes multiple sources which means that it is more likely that those gaps are filled by examples from other sources. (Frantz & Russell 2017 is not included in the current version of the database but will be included in a future release.) Relative to diachronic change, the timespan of the sources in the database is deep enough that it might contain written evidence of some recent sound changes in the language, though these may be obscured by the fact that our sources used dramatically different notation methods. Again, the concordance of tokens linked to a lemma can be inspected for changes over time between the older sources and newer sources. A useful addition in the future might be to include an InternalReconstruction field attached to each lemma. This would make the task of building correspondence sets across Algonquian and finding cognates with Blackfoot much easier.

8.3 Loss of contrast: poor evidence for innovations?

The putative phonological evidence for subgrouping involves the *loss* of word-initial *i. This aligns with a common practice in Algonquian linguistics, which is to use phonological mergers and structural loss as evidence for subgrouping (e.g. Goddard 1994). Phonological mergers and grammatical syncretisms are often hailed as excellent examples of innovations, because mergers cannot be undone (Clackson 2022: 22, Hoenigswald 1960: 151). Although mergers are an obvious sound change, it is not clear to me whether structural loss of this kind is solid evidence for subgrouping.

The best evidence for subgrouping are innovations which are particularly unusual or unlikely, because these are less likely to arise due to parallel development or drift. However, features, morphology, and phonological structure are frequently lost over time, meaning that loss of structure or loss of contrast is difficult to distinguish from drift. Without some independent criteria to distinguish the two, it seems to me that loss of structure or loss of contrast should be treated warily as evidence for subgrouping. Plausibly, there is some asymmetry between innovations which create new sounds or alter sounds versus innovations involving loss.

9 Conclusions

This paper examined a claim for a "Core Algonquian" subgropu in the Algonquian family which excludes Blackfoot. This claim is especially interesting because Blackfoot is "divergent" from the remainder of the family in several ways, and the subgroup analysis would support and explain that divergence. Ultimately, I conclude that there is no phonological evidence supporting this subgroup, and that the shared phonological feature in the "Core Algonquian" languages is better understood as a retention, with Blackfoot innovating differences in several morphophonological contexts. The innovations in Blackfoot are supported both by a synchronic analysis of root alternations and also by the historical record. This shows the importance especially of synchronic analyses and internal reconstruction when arguing for subfamily genetic relations.

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Abbreviations

3′	third person obviative
1	first person
2	second person
3	third person
AI	animate intransitive
CMD	command clause
COLLEC	collective noun
FR	Frantz & Russell (2017)
FUT	future
II	inanimate intransitive
IMP	imperative
IND	independent order
IPFV	imperfective
OBJ	object
OBV	obviative
PA	Proto-Algonquian
PAB	Proto-Algonquian-Blackfoot
P	Pentland (2023)
PL	plural
PRF	perfect
PRX	proximate
PT	particle (uninflected word)
RT	root
S	singular
SUB	subject
TA	transitive animate
TI	transitive inanimate
TI1	transitive inanimate (Class 1)
TI2	transitive inanimate (Class 2)

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Résumé

French summary goes here

Zusammenfassung

German summary goes here

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